3.5.3 Energy and ecosystems (A-level only)

Content	Opportunities for skills development
In any ecosystem, plants synthesise organic compounds from atmospheric, or aquatic, carbon dioxide. Most of the sugars synthesised by plants are used by the plant as respiratory substrates. The rest are used to make other groups of biological molecules. These biological molecules form the biomass of the plants. Biomass can be measured in terms of mass of carbon or dry mass of tissue per given area. The chemical energy store in dry biomass can be estimated using calorimetry. Gross primary production (GPP) is the chemical energy store in plant biomass, in a given area or volume. Net primary production (NPP) is the chemical energy store in plant biomass after respiratory losses to the environment have been taken into account, ie $NPP = GPP - R$ where GPP represents gross production and R represents respiratory losses to the environment. This net primary production is available for plant growth and reproduction. It is also available to other trophic levels in the ecosystem, such as herbivores and decomposers. The net production of consumers (N), such as animals, can be calculated as: $N = I - (F + R)$ where I represents the chemical energy store in ingested food, F represents the chemical energy lost to the environment in faeces and urine and R represents the respiratory losses to the environment. Primary and secondary productivity is the rate of primary or secondary production, respectively. It is measured as biomass in a given area in a given time eg kJ ha^{-1} year^{-1}. **Students should be able to** appreciate the ways in which production is affected by farming practices designed to increase the efficiency of energy transfer by: • simplifying food webs to reduce energy losses to non-human food chains • reducing respiratory losses within a human food chain.	**MS 0.1** Students could be given data from which to calculate gross primary production and to derive the appropriate units. **AT a** Students could carry out investigations to find the dry mass of plant samples or the energy released by samples of plant biomass. **MS 2.4** Students could be given data from which to calculate: • the net productivity of producers or consumers from given data • the efficiency of energy transfers within ecosystems. **MS 0.3** Students could be given data from which to calculate percentage yields.